

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Owned 7-deployment Node.js backend: real-time event pipelines, Temporal.io workflow orchestration, RBAC, and multi-tenant data isolation at production scale. - Built backend: PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry; 394 commits across 7 production deployments.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-based genomic data platform on Google Cloud Run (GCP) with JBrowse 2 genome browser; 100+ researchers, 500GB datasets. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation using OCR, NLP, and ML classification pipelines; improved processing accuracy by 25%. - Automated deployments on AWS and GCP; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Demand Paging	Xinu OS – Systems Programming	2024
- Engineered 8-store virtual memory system in C for Xinu OS using Second-Chance page replacement; reduced page faults by 25% and memory overhead by 15%. - Added 7+ kernel system calls with inverted page tables and new fault handler; 15% memory performance gain.		
VectorLift	Semantic Search Engine	2026
- Semantic search in Python on MS MARCO: BM25, dense bi-encoder (in-batch negatives), cross-encoder re-ranker. - Scored NDCG@10, MRR@10, MAP; FastAPI serving with per-stage latency; Kafka-driven Qdrant corpus indexing.		
Inventory Reservation Service	Backend API	2026
- Java 21/Spring Boot REST API: PostgreSQL row-level locking (SELECT FOR UPDATE), idempotent POSTs, RFC 9457 errors, scheduled expiry across multi-SKU reservations. - Testcontainers: two concurrent POSTs produce exactly one 200, one 409; Flyway, Docker, GitHub Actions CI.		
LLM Eval Harness	Independent Project	2025
- Vendor-neutral benchmark comparing Claude, GPT-4o, Gemini across 2,000 examples and 3 task types in Python. - Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.		

Languages and Technologies

- Languages:** C, Python, Java, TypeScript, JavaScript, SQL, Bash
- Systems & Infra:** Docker, Kubernetes, AWS, GCP, Kafka, Spark, Hadoop, Redis, PostgreSQL
- Data & ML:** NumPy, Pandas, PyTorch, LLM evaluation, RAG, LangChain, FastAPI, DuckDB
- Frameworks:** Spring Boot, Node.js, NestJS, Temporal.io, GitHub Actions, OpenTelemetry, Grafana

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.